

# torch.compiler.set\_enable\_guard\_collect

[https://pytorch.org/docs/stable/generated/torch.compiler.set\\_enable\\_guard\\_collect](https://pytorch.org/docs/stable/generated/torch.compiler.set_enable_guard_collect)

## Description:

Enables use of collectives during guard evaluation to synchronize behavior across ranks. This is expensive: we have to issue a collective every time we enter a compiled code region, even if no rank actually would need to compile. This can help prevent NCCL hangs by ensuring that we never have a situation where one rank starts recompiling while other ranks don't compile; it is especially useful in conjunction with `enable_compiler_collectives` where such a situation would immediately cause a hang (as it is necessary for all ranks to compile at the same time to run compiler collectives). Like compiler collectives, you can only run this on SPMD programs; you will hang otherwise. Note that a guard collective is only issued if there is any compiled code to guard on; if this is the first time we encounter a frame or the frame is skipped, we don't issue collectives. Returns the previous setting of enabled.

## Function Signature

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```
torch.compiler.set_enable_guard_collectives(enabled)  
[source]#
```

## Detailed Documentation

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### Documentation

Enables use of collectives during guard evaluation to synchronize behavior across ranks. This is expensive: we have to issue a collective every time we enter a compiled code region, even if no rank actually would need to compile. This can help prevent NCCL hangs by ensuring that we never have a situation where one rank starts recompiling while other ranks don't compile; it is especially useful in conjunction with `enable_compiler_collectives` where such a situation would immediately cause a hang (as it is necessary for all ranks to compile at the same time to run compiler collectives). Like compiler collectives, you can only run this on SPMD programs; you will hang otherwise. Note that a guard collective is only issued if there is any compiled code to guard on; if this the first time we encounter a frame or the frame is skipped, we don't issue collectives.

Returns the previous setting of `enabled`.